Session - 1: The UAV Technology Landscape: PX4, ArduPilot, and Beyond

Agenda

Introduction to PX4 and ArduPilot

India Towards Indigenization in UAV Technology

How we can become a solution architect in UAV

UAV - Practical Approach when selecting the components

System Architecture & Components

Development Setup on Ubuntu for Ardupilot

QGroundControl Simulation Demo

Case Study

Journey So Far

🎓 **Education:** M.Tech in Mechatronics

🧩 **Specialization:** Embedded systems, UAV autonomous systems, ROS2, Ground Control Station, Solution Architect

- 🛠️ **Projects:**
  - Integrated **PX4 and ArduPilot** on Pixhawk boards for autonomous drone testing
  - Developed custom **offboard control scripts** using **pymavlink and MAVROS**
  - Worked with **QGroundControl** for flight mission setup and SITL testing
  - Ardupilot Porting on STM32 Boards
- 🌍 **Industry Exposure:**
  - Worked on UAV system integration and autopilot customization
  - Research on **encrypted firmware bootloaders** for secure UAV updates
  - Solution Architect
  - Robotic Arms and Mobile Robots
- 🚀 **Current Focus:** Bridging **academic learning** with **industrial UAV practices** and New Inventions in a field of Robotics
- Worked for Indian Army, RRU, Indian Airforce, DRDO

Welcome & Objectives

✈️ *Goal:* Understand PX4, ArduPilot, and how to connect with QGroundControl

Outcomes:

How to Create a thought process of Solution Architect

UAV Solution Architecture Layers

1. **Mission Layer** – Use case, KPIs, constraints
2. **UAV Platform** – Airframe, propulsion, payload
3. **Avionics & Autopilot** – PX4/ArduPilot, sensors
4. **Communication** – RF, LTE, SATCOM, mesh
5. **Edge & Cloud** – AI inference, data processing
6. **Ground Segment** – GCS, dashboards, APIs
7. **Operations & Compliance** – DGCA, safety, SOPs

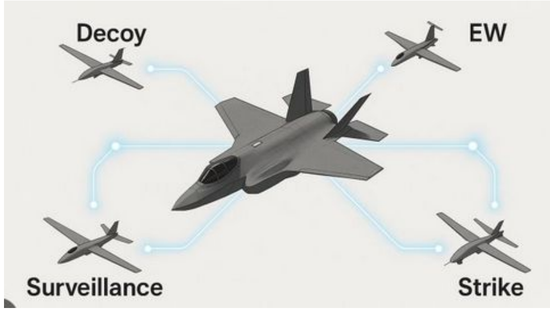
Autopilot & Flight Stack Understanding

- Sensor fusion (IMU, GPS, Compass, Barometer)
- Control loops (PID)
- RCIN vs RCOUT
- Failsafes & redundancy
- Logs & diagnostics (MAVLink, DataFlash)
- Firmware customization & parameters

💡 **Architect thinks:**
“How will this behave in failure, scale, and edge cases?”

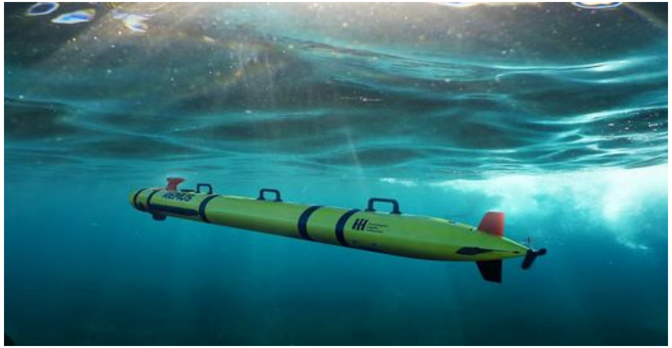
Emerging Industrial Trends in Autonomous Platforms

CCA



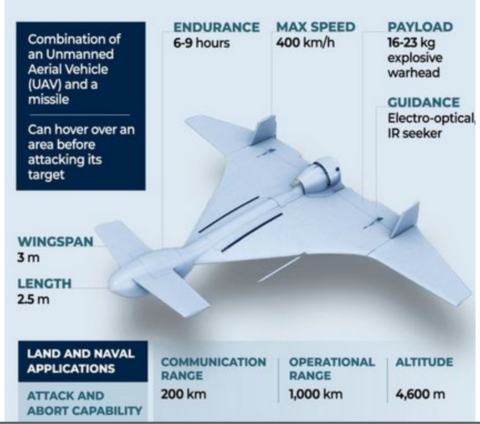
Loitering Munitions

HAPS

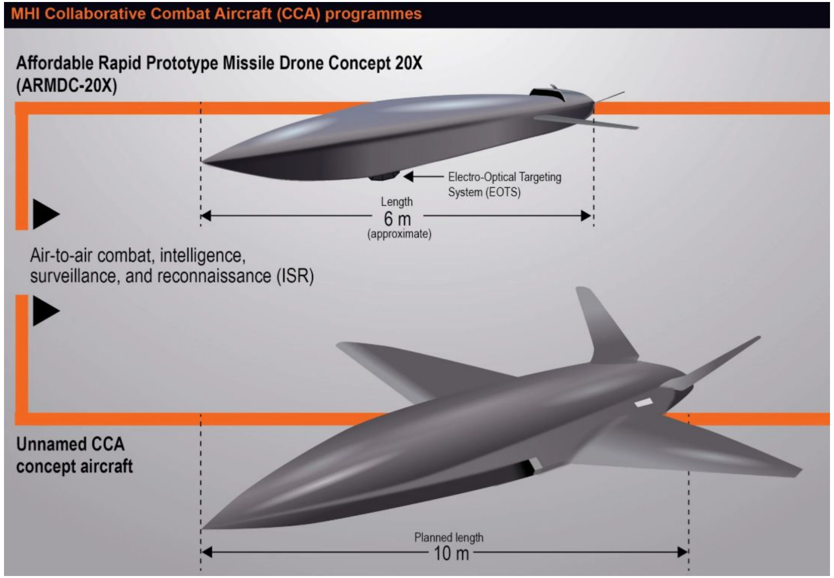


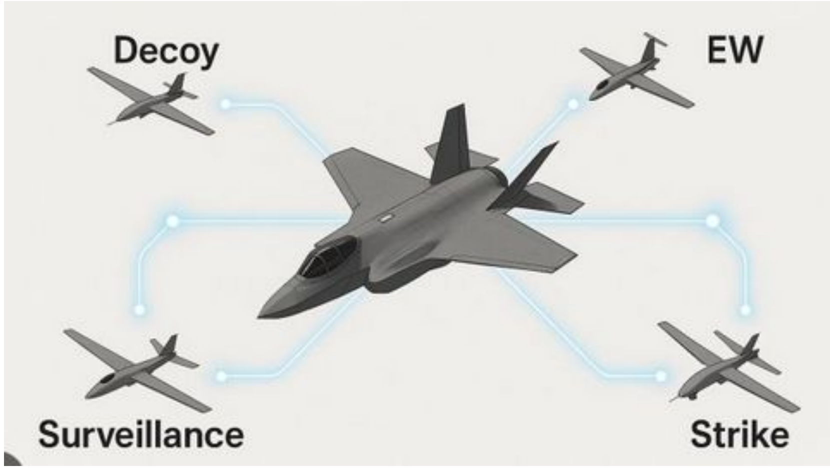
UUV

🚀 **New Directions Beyond Traditional Drones**

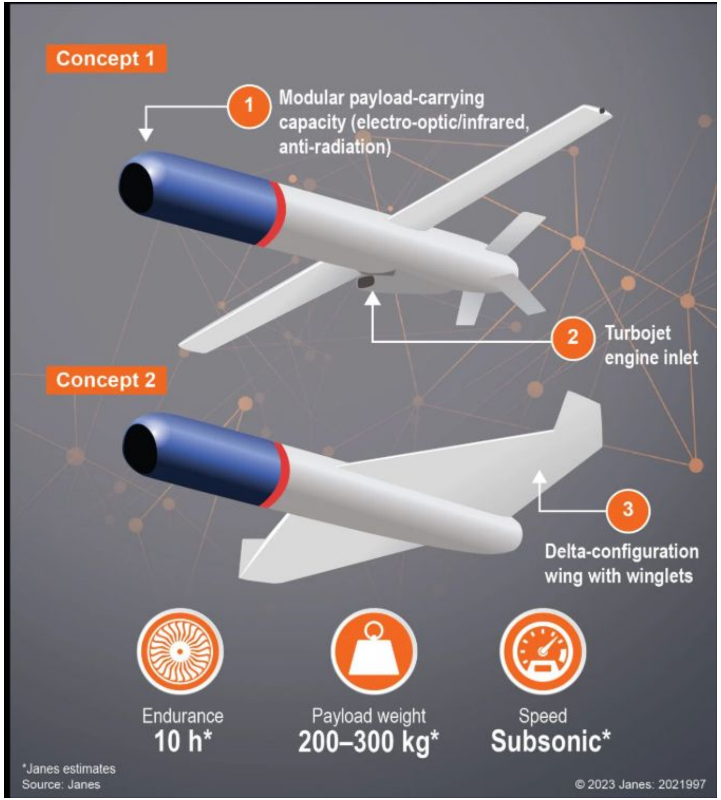


CCA UAV – Collaborative Combat Aircraft





Loitering Munitions



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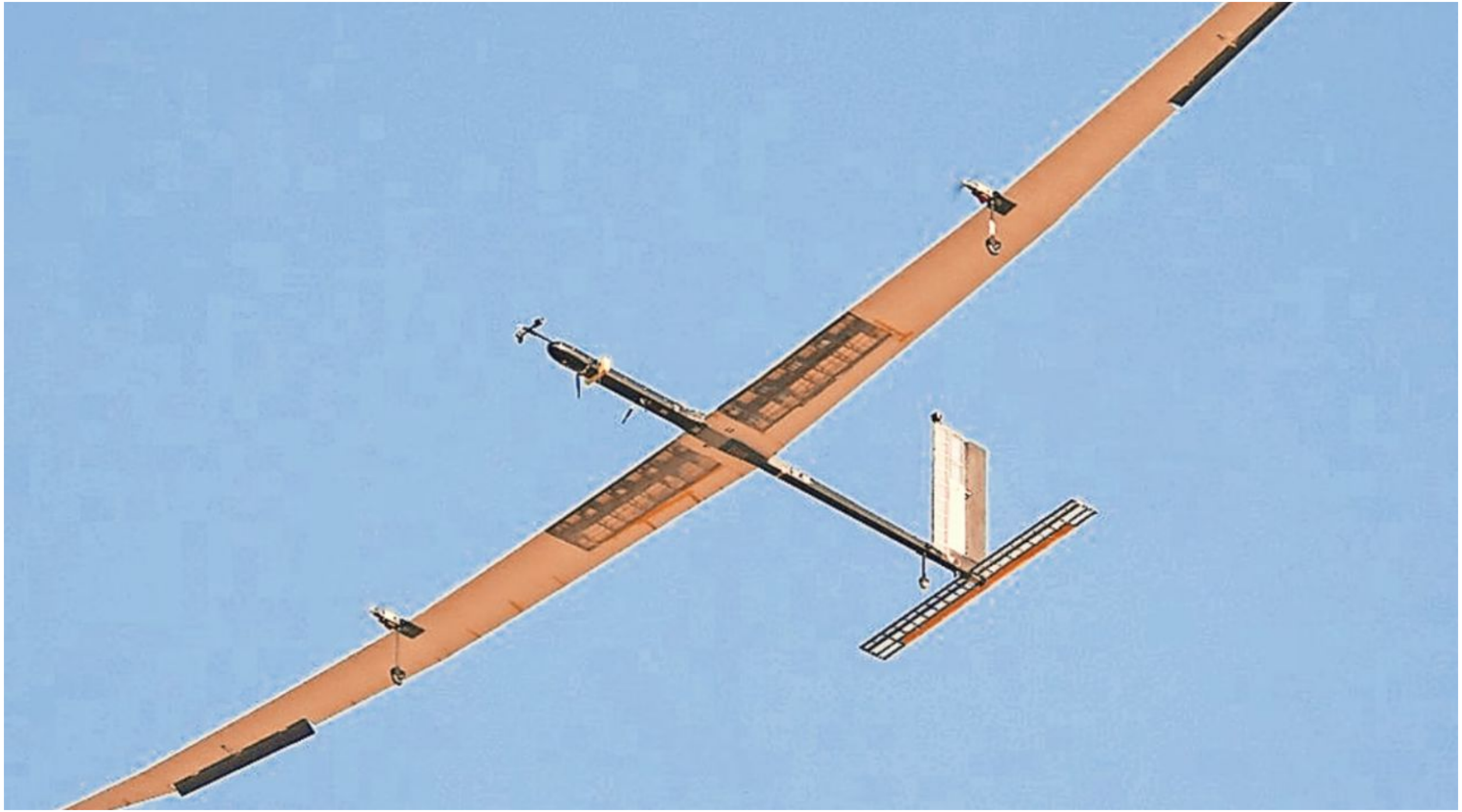

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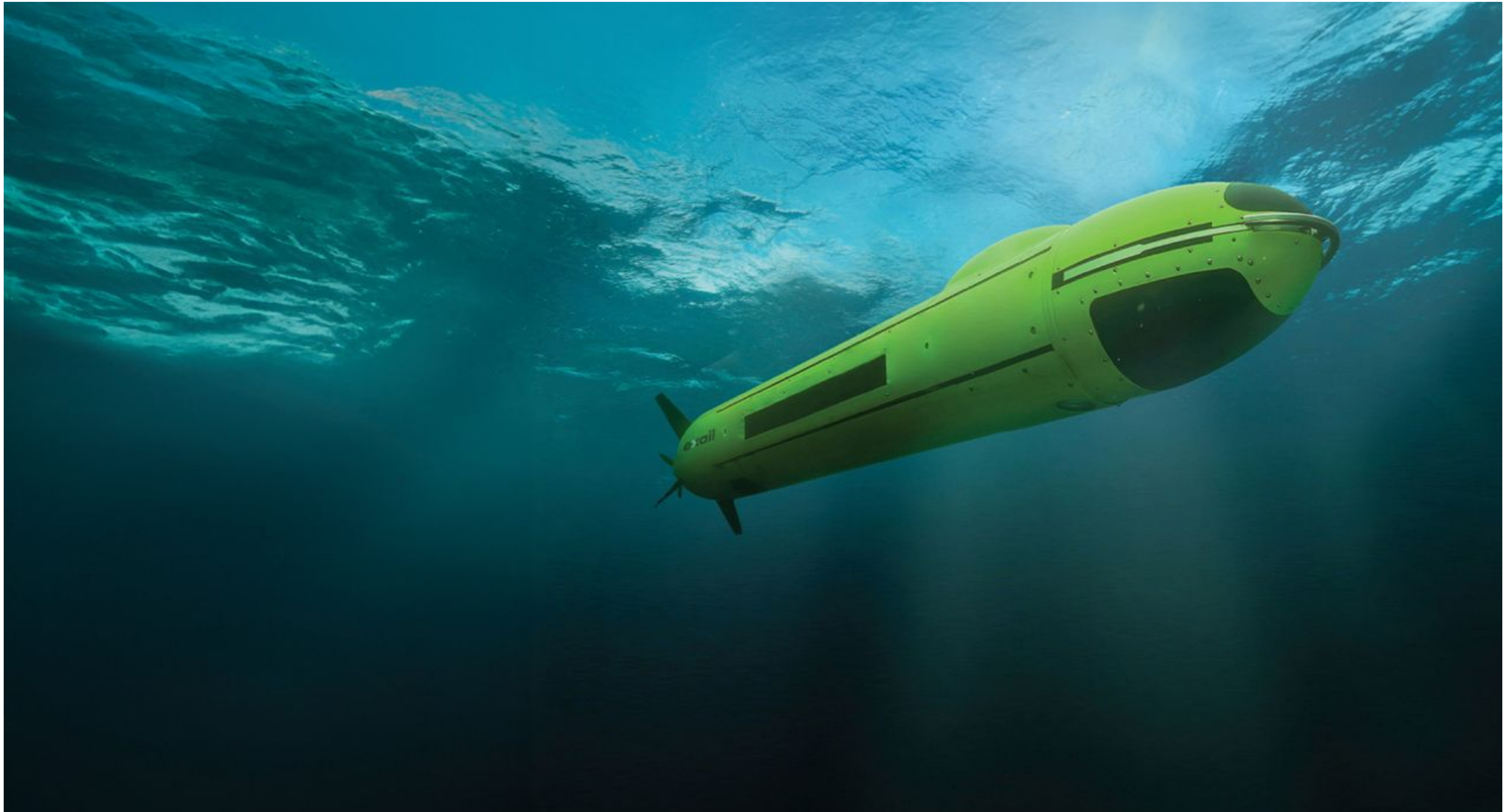

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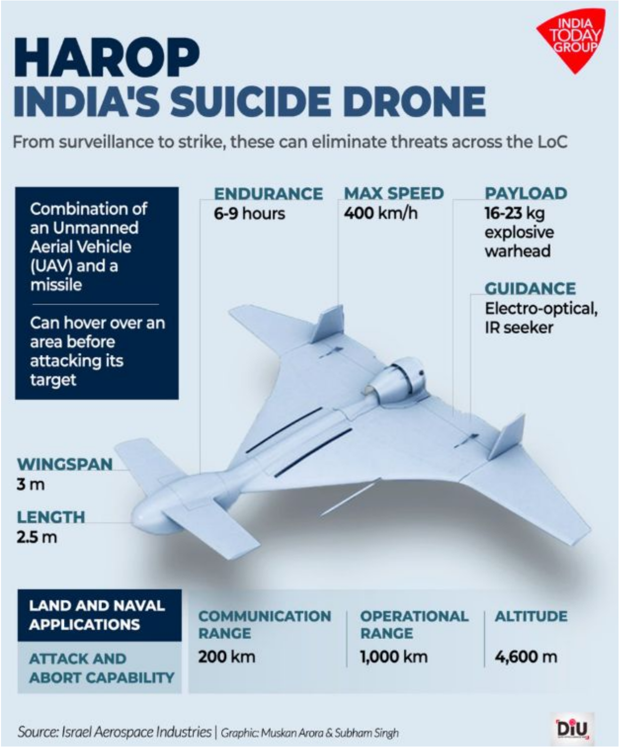

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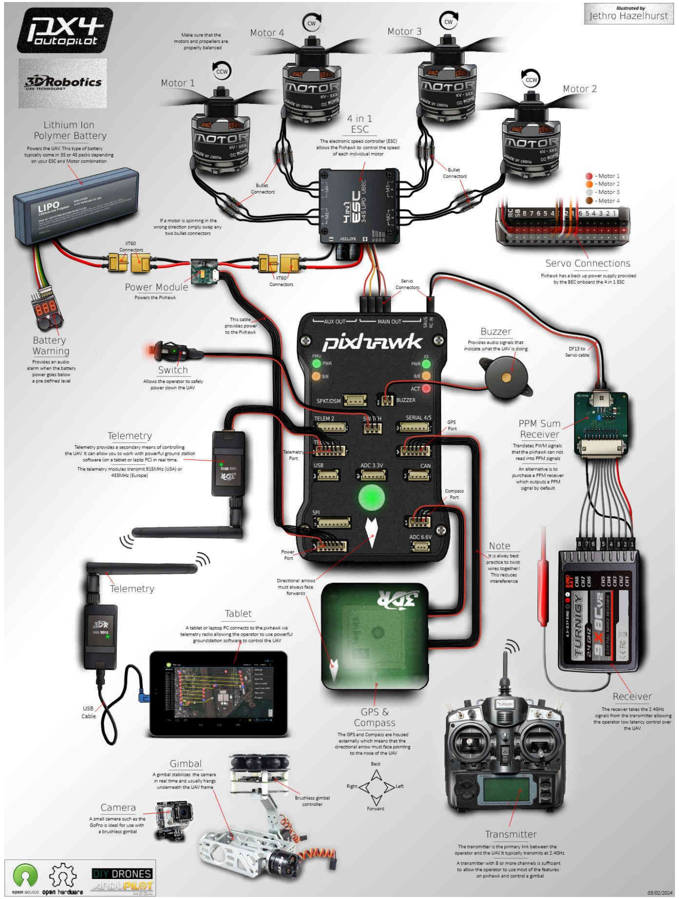
HAROP
INDIA'S SUICIDE DRONE
From surveillance to strike, these can eliminate threats across the LoC
Combination of an Unmanned Aerial Vehicle (UAV) and a missile
Can hover over an area before attacking its target
ENDURANCE
6-9 hours
MAX SPEED
400 km/h
PAYLOAD
16-23 kg explosive warhead
GUIDANCE
Electro-optical, IR seeker
WINGSPAN
3 m
LENGTH
2.5 m
LAND AND NAVAL APPLICATIONS
ATTACK AND ABORT CAPABILITY
COMMUNICATION RANGE
200 km
OPERATIONAL RANGE
1,000 km
ALTITUDE
4,600 m
Source: Israel Aerospace Industries | Graphic: Muskan Arora & Subham Singh
DIU

Drone Basic Components

Category	Component	Function
🛠️ **Structure**	**Frame**	Holds all parts together; provides shape and stability.
⚙️ **Propulsion System**	**Motors, Propellers, ESCs**	Generate thrust and control drone movement.
🔋 **Power System**	**Battery, Power Distribution Board (PDB)**	Supplies and distributes electrical power to components.
🧠 **Flight Controller**	**Pixhawk, Cube, Navio2, etc.**	The “brain” — processes sensor data and stabilizes flight.
📡 **Sensors**	**IMU, GPS, Compass, Barometer, LiDAR**	Provide position, orientation, and environmental data.
📶 **Communication System**	**Radio Receiver, Telemetry Module**	Enables control and data exchange between drone and ground.
💻 **Ground Control Station (GCS)**	**QGroundControl, Mission Planner**	Interface for monitoring, mission planning, and control.
🎯 **Payload**	**Camera, Sprayer, Delivery Box, etc.**	Mission-specific component for data capture or delivery.
🧩 **Software Stack**	**PX4, ArduPilot, MAVLink**	Autopilot firmware and communication software.



Motor – Propeller – Battery Correlation for UAV Selection

Architect’s Golden Rule

Battery feeds the motor, motor turns the prop, prop lifts the mission
Weakest link defines system failure.

2. Motor Selection

- **KV Rating**
  - Low KV → Large prop, high torque, efficiency
  - High KV → Small prop, high RPM, speed
- **Max Continuous Current**
- **Max Power (W)**
- **Motor torque capability**

3. Propeller Selection

- **Diameter** → Thrust generation
- **Pitch** → Airspeed & current draw
- Larger prop = more thrust, **higher torque demand**
- Prop must match **motor torque curve**

4. Battery Selection

- **Voltage (S count)** → Sets motor RPM
- **Capacity (mAh)** → Flight time
- **C-rating** → Current supply

$$I_{required} \leq Capacity \times C$$

5. System Correlation (Key Equation)

$$Power = Voltage \times Current$$

- Prop ↑ → Current ↑
- KV ↑ → RPM ↑ → Current ↑
- Battery voltage ↑ → Current ↓ (for same power)

6. Safety Margins (Architect Rule)

- Motor operating at **≤ 70% max current**
- ESC rating **≥ 30% above max current**
- Battery discharge **≤ 80% capacity**
- Prop RPM **below structural limit**

7. Final Validation

- Bench thrust test
- ESC temperature
- Battery voltage sag
- Hover throttle (ideal: **40–60%**)

Band & Frequency Hopping in UAV Communication

What is Frequency Hopping?

Frequency Hopping Spread Spectrum (FHSS) is a technique where the UAV radio **rapidly switches (hops)** between multiple frequencies within a defined band, following a **pseudo-random sequence** known to both transmitter and receiver.

What is Band Hopping?

- **Band hopping** = switching between **different frequency bands**
 (e.g., 900 MHz ↔ 2.4 GHz)

- **Frequency hopping** = switching **within the same band**
 (e.g., multiple channels inside 2.4 GHz)

Why Frequency Hopping is Critical in UAVs

a) Anti-Interference

- Avoids Wi-Fi, other drones, EMI
- If one channel is noisy → next hop is clean

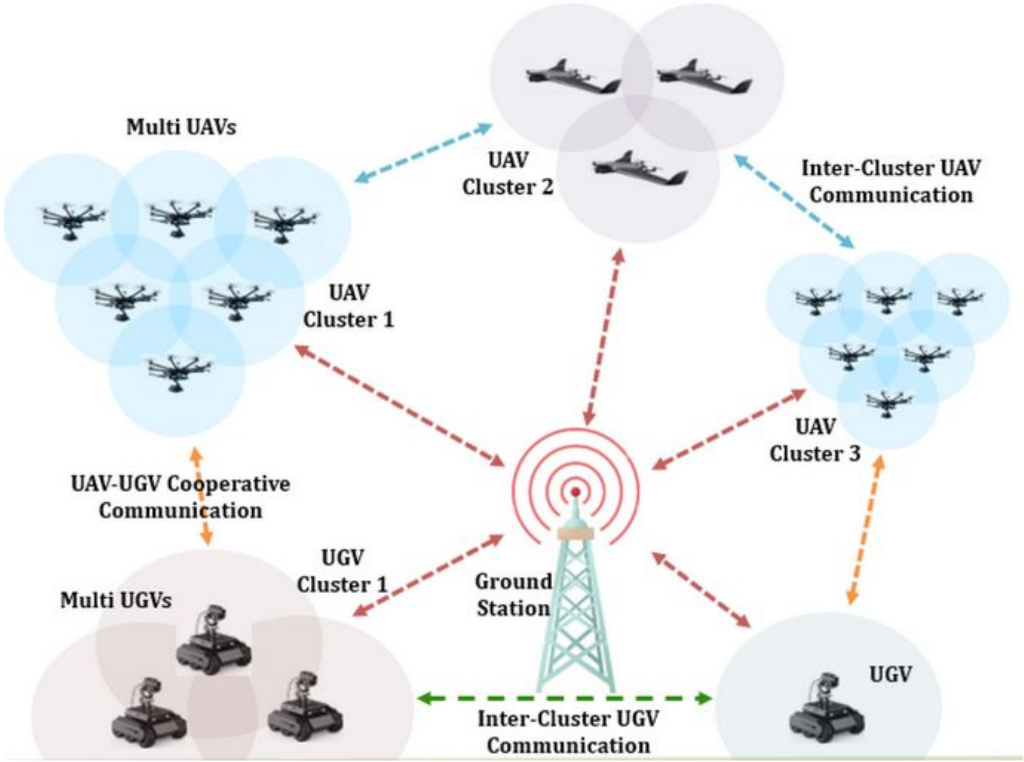
b) Anti-Jamming

- Hard for an attacker to jam all hopping frequencies
- Used heavily in **defense and ISR UAVs**

c) Link Reliability

- Reduces packet loss
- Improves control stability and telemetry integrity

Collaborative Mesh Network



Indigenization in UAV Technology: Building Self-Reliant Drone Ecosystems in India

"By mastering open-source platforms like PX4 and ArduPilot, Indian engineers can **customize, innovate, and localize** UAV technologies — making India not just a user, but a **global leader in autonomous systems.**"

Domain	Goal	Indian Opportunity
Flight Controllers	Develop PX4/ArduPilot compatible Indian hardware (Pixhawk alternatives)	HAL, Bharat Electronics, IIT start-ups, Botlab Dynamics, ZeroDrag
Sensors & IMUs	Replace imported GPS, barometers, and IMUs	DRDO labs, MSMEs developing MEMS sensors
Communication Modules	Indigenous telemetry, RF, and 4G/5G modules	Local RF design and manufacturing
Power Systems	Indian-made Li-ion cells and BMS systems	ISRO, private energy startups
Software & Autonomy	Build AI, computer vision, and PX4/ArduPilot forks in India	Open-source collaboration, academia-industry R&D

Explaining PX4 and ArduPilot in Layman's Terms

- **The Researcher's Brain**

  PX4 is like a **flexible and modern autopilot system** — designed for people who want to **experiment, innovate, and build new types of drones**.
  It's highly **modular** easy to **integrate** sensors and **actuators**

- **The Industry Veteran**

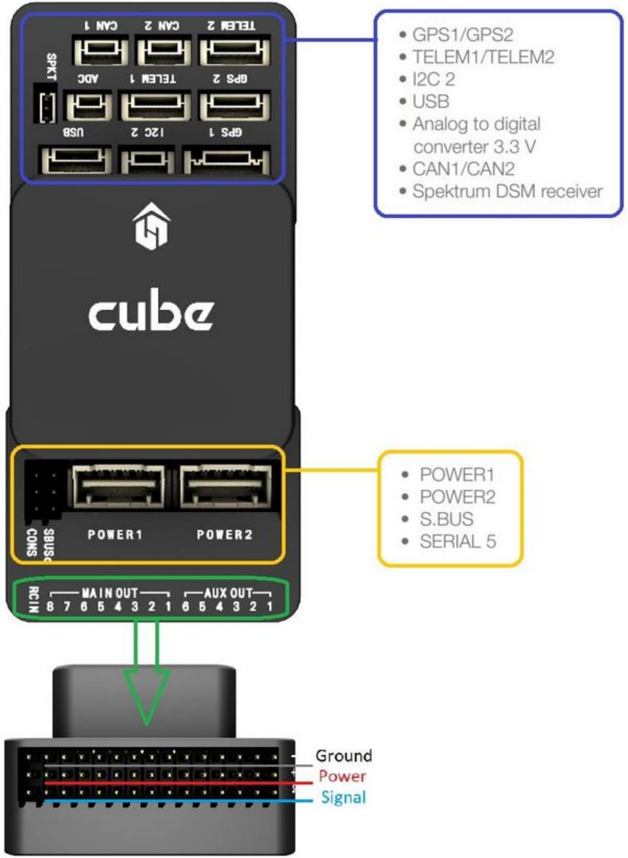
  ArduPilot is like a **seasoned pilot** — it's older, very experienced, and can handle **many types of vehicles**, not just drones.
  It's used in professional and commercial drones that need **reliability and stability** more than constant innovation.

Simple Metaphor to End With

- PX4 and ArduPilot are like the **Android and iOS of drones**.
- PX4 is more open and modular
- ArduPilot is more mature and widely used.

Hardware Anatomy

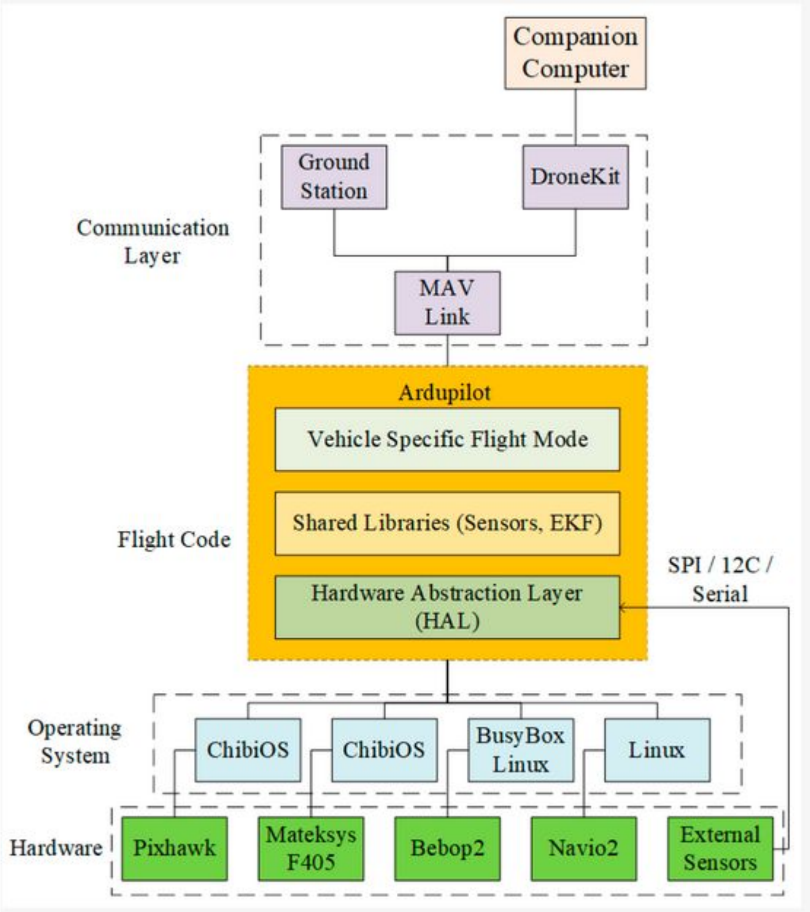
- **Sensors:** (all connected via SPI)
  - **Accelerometer:** (3) ICM42688p, ICM20948, ICM20649
  - **Gyroscope:** (3) ICM42688p, ICM20948, ICM20649
  - **Compass:** (1) ICM20948
  - **Barometric Pressure Sensor:** (2) MS5611



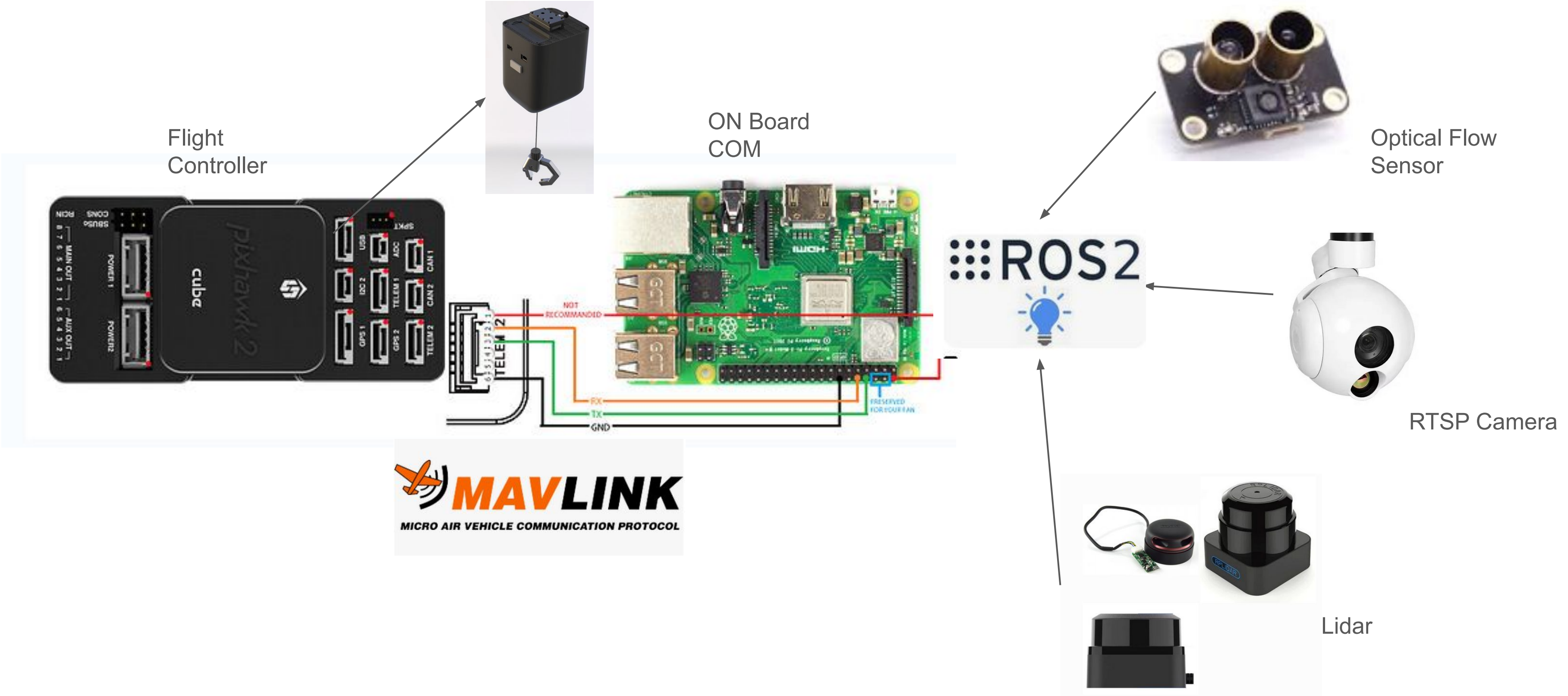
Software Anatomy

Main Layers:

- 🧩 **HAL (Hardware Abstraction Layer)** – Bridge between Hardware and Software
- ⚙️ **Drivers & Libraries** – Sensor interfaces and core functions
- 🛫 **Vehicle Firmware** – ArduCopter, ArduPlane, ArduRover, etc.
- 🕹️ **Control Loops** – Stabilize and navigate using PID algorithms
- 📡 **MAVLink Interface** – Communication with GCS (QGC, Mission Planner)
- 💾 **Parameter & Logging System** – Save settings and flight data



Make It More Intelligent



Case Study: 15 kg MTOW Multi-Rotor UAV System

Objective:
Design a **15 kg MTOW multi-rotor UAV** capable of carrying a **useful payload** with safe, reliable, and compliant operation.

Primary Use Cases

- Industrial inspection
- Surveillance & ISR
- Mapping & data acquisition
- Logistics (light cargo)

Parameter Requirement

- MTOW = **15 kg**
- Payload = 3–4 kg
- Endurance = 30–40 minutes
- Range = 10–15 km
- Flight Mode = Autonomous + Manual
- Redundancy = Medium (civil/industrial)
- Environment = Wind ≤ 8 m/s

Install Ardupilot SITL and QGroundControl

SITL - https://ardupilot.org/dev/docs/setting-up-sitl-on-linux.html

QGroundControl - https://qgroundcontrol.com/